Government Spending Shocks and Consumption Inequality*

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Abstract

Fiscal policy is often anticipated before implementation, yet little is known about how fiscal foresight shapes inequality. We study the distributional effects of government spending shocks, distinguishing between anticipated (*news*) and unanticipated (*surprise*) components. Using the identification strategy of Forni and Gambetti (2016), we show that positive surprise shocks reduce U.S. consumption inequality, whereas positive news shocks increase it. A borrower - saver New Keynesian model rationalises these findings and highlights the role of monetary - fiscal regimes: in a monetary-led regime, news and surprise shocks have opposite distributional effects, while in a fiscally-led regime both reduce inequality. Our results demonstrate that the distributional impact of fiscal policy depends critically on fiscal foresight and the prevailing policy regime.

Keywords: consumption inequality, fiscal policy, news shocks, monetary-fiscal interaction, TANK model

JEL Codes: C32, E21, E61, E62, E63

- Subset of variables for IRFs in main section: Both G, Y, C, consumption ineq
- Try different anticipation horizons
- Discuss with Alistair (maybe send draft?)
- Motivation: could it be useful to frame it in the current environment fiscal debt and concerns about fiscal dominance?

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1 Introduction (F+R+V)

Anticipation is a central feature of fiscal policy. Changes in government spending and taxation are typically subject to legislative and implementation lags, and are often announced well before they take effect. However, due to the presence of fiscal foresight, private agents adjust their expectations and behaviour in advance of the actual policy change (see Leeper et al. (2012, 2013), among others). A growing literature shows that these anticipation effects are quantitatively important and can even dominate implementation dynamics. Mertens and Ravn (2011) find that anticipated tax cuts generate contractions prior to implementation and expansions once enacted. Similarly, Ascari et al. (2023) show that anticipated government spending shocks can be either contractionary or expansionary at announcement, depending on the monetary-fiscal regime. Fiscal shocks are therefore inherently multi-dimensional, comprising both *surprise* and *news* components that propagate through distinct channels.

While fiscal anticipation is well understood at the aggregate level, its distributional implications remain underexplored. Existing evidence shows that positive government spending surprise shocks reduce consumption inequality (e.g., Anderson et al. (2016a)), a result typically rationalised in two-agent New Keynesian models where Ricardian households smooth consumption and hand-to-mouth households increase spending in response to higher income, as in Galí et al. (2007a). However, if fiscal policy is largely anticipated, focusing only on surprise shocks provides an incomplete picture. Accounting separately for news and surprise shocks is therefore essential to understand the distributional effects of fiscal policy.

This paper examines the impact of government spending shocks on U.S. consumption inequality, distinguishing between anticipated and unanticipated shocks. Using microdata from the Consumer Expenditure Survey (CEX), we construct measures of consumption inequality following Coibion et al. (2017a). Our results reveal starkly different distributional effects. Consistent with previous evidence, a positive surprise shock to government spending reduces consumption inequality. In contrast, a positive news shock increases inequality. To rationalise these findings, we use the borrower-saver New Keynesian framework of Bilbiie et al. (2013), under two alternative monetary-fiscal regimes. In the monetary-led regime, the central bank follows the Taylor Principle, and the government adheres to fiscal discipline. In the fiscally-led regime, fiscal policy is exogenous, and the central bank accommodates it by passively setting interest rates. We show that the regimes yield contrasting distributional effects of fiscal surprise and news shock; while surprise shocks reduce consumption inequality in the monetary regime and news shocks increase consumption inequality, both shocks reduce consumption inequality in the fiscal regime.

The novelty of our results lies in showing the contrasting distributional implications of government spending news and surprise shocks, contingent on the (monetary/fiscal) policy regime in place. The identification of fiscal shocks relies on the strategy proposed by Forni

and Gambetti (2016), which combines a recursive identification scheme àla Blanchard and Perotti (2002) with revisions in Survey of Professional Forecasters (SPF) expectations to recover orthogonal surprise and news shocks to government spending. Our work is the first to employ it to examine the impact on consumption inequality

Our work relates to two broad strands of literature. First, it connects to the empirical literature on the distributional effects of government spending. Anderson et al. (2016a) identify unanticipated spending shocks using the forecast-error measure of Ramey (2011a) and show, using CEX data, that surprise spending increases reduce consumption inequality. Ma (2019a) shows that in response to unanticipated government spending increases, consumption increases for the poor households, whereas it decreases for the rich. Using CEX data, Dhital et al. (2023) show that government spending shocks lower income, disposable income, consumption and expenditure inequality. Evidence on anticipated shocks is more limited. De Giorgi and Gambetti (2012) find that news shocks reduce inequality, but their analysis does not jointly identify orthogonal surprise and news shocks and focuses exclusively on non-durable consumption. Second, our paper contributes to the growing literature on fiscal policy in New Keynesian frameworks, focusing on anticipation and monetary-fiscal interactions, building on existing work by Ascari et al. (2023) and Mertens and Ravn (2011). [1]

The rest of the paper is organised as follows. Section 2 describes our empirical analysis, including the data and the identification strategy used, and presents the response of consumption inequality to government spending news and surprise shocks. Section 3 describes the model and provides an analytical illustration of our main result as well as the simulated responses in both monetary-led and fiscal-led regimes. Finally, Section 4 concludes. Robustness checks are presented in the Appendix.

Should we have intro and literature review in a unique section?

Discussion about fiscal foresight, why disentangling fiscal and news shocks is paramount (it should be centered on Leeper et al., 2013). Mention empirical works that address fiscal foresight in the identification of surprise government spending shocks (for instance, Ramey, 2011b, but the list should be way longer); finally, introduce Forni and Gambetti (2016), that develops the identification scheme we use in the paper.

SOMETHING ALONG THESE LINES: Fiscal anticipation entails an important difficulty for VAR analysis. As shown by Leeper et al. (2013), if the VAR does not contain enough information, the underlying structural MA representation of the variables in the VAR is not invertible, or 'non-fundamental'. In this case, the VAR is misspecified and can lead to misleading results. Evidence of non-fundamentalness for a standard closed economy VAR specification is provided by Ramey (2011b). To avoid that informational deficiency affects the

[1] Add papers on FP in NK frameworks, check with everyone on references to add

results of our empirical analysis, we follow the approach introduced by Forni and Gambetti (2014). A VAR is defined as 'informationally sufficient' if, and only if, the structural shocks are not Granger caused by the common factors driving macroeconomic variables. If a set of variables is not sufficient, the solution consists in estimating a Factor Augmented VAR model (FAVAR), where the original set of variables is enlarged with the principal components.

2 Empirical analysis

2.1 Data

2.1.1 Macroeconomic variables

Our empirical analysis is based on U.S. quarterly data. We employ a medium-scale model which includes eight macroeconomic and financial variables: (i) real federal government consumption expenditures and gross investment; (ii) real gross domestic product (GDP); (iii) the federal budget surplus as a share of nominal GDP; (iv) the 10-year Treasury constant maturity rate; (v) the real exchange rate; (vi) the federal funds rate; (vii) corporate profits; (viii) consumer confidence. Data sources are detailed in Appendix A. Real variables are obtained by deflating nominal values with the GDP deflator. In Appendix B, we run the test proposed by Forni and Gambetti (2014), which indicates that our model specification is informationally sufficient and thus not affected by non-fundamentalness. In Appendix E, we address informational deficiency with a small-scale VAR augmented with common factors from a large macroeconomic dataset (FAVAR). This alternative approach delivers very similar results.

2.1.2 Household-level data

Consumption inequality is measured using the indexes in Coibion et al. (2017b), recently extended by Mangiante and Meichtry (2025) until December 2019. They are obtained from the Consumer Expenditure Survey (CEX), which is the most granular source of information on U.S. household consumption. The CEX is a monthly rotating panel in which households are interviewed once per quarter, for up to five consecutive quarters. In each round, respondents report expenditure for the three months preceding the interview. Household consumption is defined as the sum of non-durables, services, and selected durable goods, such as household appliances, televisions, and furniture. Expenditures on large durables, such as house and car purchases, is excluded, as they resemble a form of investment rather than consumption. Our main measure of consumption inequality is the cross-sectional standard deviation of household-level real consumption, denoted by c_t^{sd}. In the Appendix, we show that our results are robust to using alternative indicators.

2.1.3 Fiscal news

To identify news shocks to government spending, we follow the identification strategy of Forni and Gambetti (2016), which hinges on data from the Survey of Professional Forecasters (SPF, henceforth). This is a quarterly survey conducted by the Federal Reserve Bank of Philadelphia, which collects expectations from professional economists about key U.S. macroeconomic series, such as GDP growth, inflation, unemployment, and government spending. Forni and Gambetti (2016) exploit SPF forecasts for government spending growth to derive a series that is affected contemporaneously by the fiscal news shock. This variable is later included into a VAR, ordered second after government spending. The surprise and news shocks are then identified as the first and second shock of the Cholesky decomposition, respectively.

Our paper relies on two alternative news series. The first is defined as

$$F_t(1, H) = \sum_{h=1}^{H} \hat{E}_t^I \Delta g_{t+h}, \tag{2.1}$$

where $\hat{E}_t^I \Delta g_{t+h}$ is the h-quarter-ahead median SPF forecast for government spending growth. The second one is instead

$$n_t(1, H) = F_t(1, H) - F_{t-1}(2, H + 1). \tag{2.2}$$

Following Forni and Gambetti (2016), we employ $F_t(1, H)$ in our baseline model, with $H = 4$. Our findings are robust to using the measure $n_t(1, H)$, as detailed in Appendix D.1.

2.2 Results

This section presents the results from our medium-scale VAR model. It consists of the variables described in 2.1.1, along with $F_t(1, 4)$, ordered second after government spending, and c_t^{sd}. The sample goes from 1981:IV to 2019:IV; the start start date is dictated by the availability of SPF forecasts and the sample ends before the Covid-19 outbreak. The model is estimated using Bayesian techniques, specifically using Jeffreys priors, four lags and an intercept. Surprise and news shocks to government spending are identified as the first two shocks in the Cholesky decomposition (Forni and Gambetti, 2016).

Our main results are Figures 1 and 2. The responses are broadly consistent with those obtained by Forni and Gambetti (2016). Both shocks are expansionary, with the news shock leading to a greater increase in output. The starkest difference concerns the transmission to government spending and long-term interest rate. After a surprise shock, spending increases on impact but then gradually decreases, with bond yields moving downward. On the other hand, government spending rises progressively in response to a news shock, reaching its maximum after about two years. In this case, spending reversal occurs with a significant delay, resulting

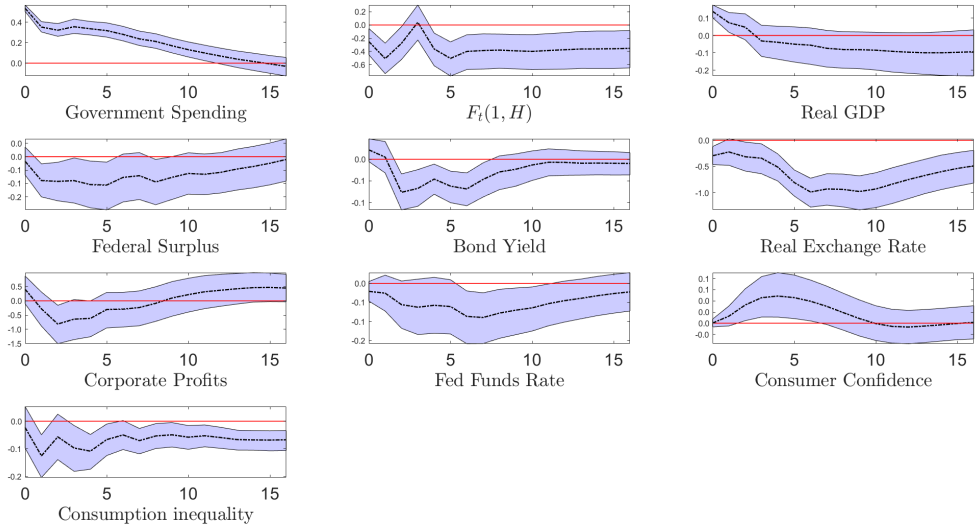


Figure 1: IRFs of macroeconomic variables to a surprise shock to government spending.
Notes: solid line is the point-wise posterior median response. Shaded bands are 68% equal-tailed credibility intervals.

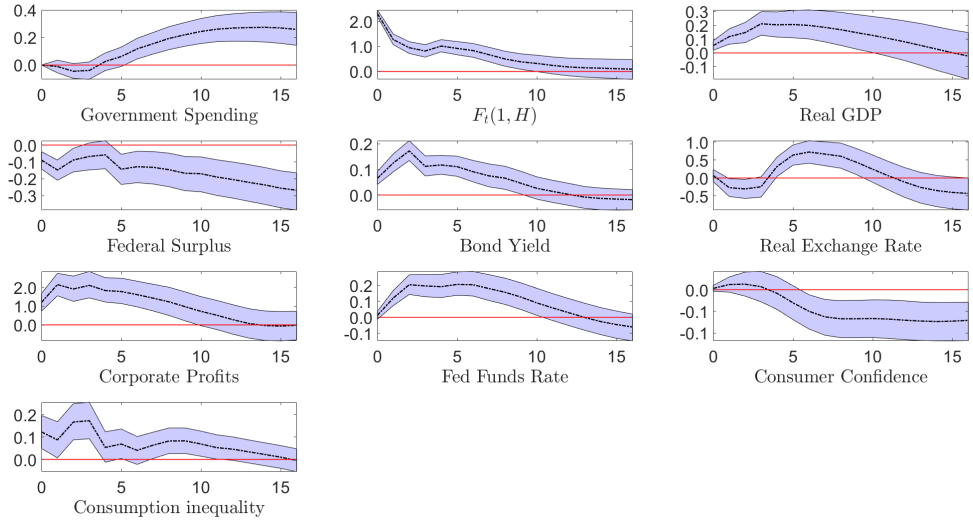


Figure 2: IRFs of macroeconomic variables to a news shock to government spending.
Notes: solid line is the point-wise posterior median response. Shaded bands are 68% equal-tailed credibility intervals.

in a sharp increase in the long-term interest rate. Notably, the surprise shock generates a decline in consumption inequality. This is a well-documented result (e.g., Anderson et al., 2016b; Ma, 2019b), which aligns with predictions from TANK models with unconstrained and hand-to-mouth agents (e.g. Galí et al., 2007b). A news shock, on the contrary, increases

consumption inequality. This is a new finding in the empirical literature on government spending, suggesting that fiscal anticipation plays an important role in shaping the response of the consumption distribution. In Section 3, we rationalize this finding through the lenses of a TANK model with borrowers and savers.

In Appendix D, we discuss various departures from our baseline model, which corroborate the results presented in this section. In particular, the following robustness checks are considered: (i) replacing $F_t(1,4)$ with $n_t(1,4)$; (ii) replacing c_t^{sd} with the Gini coefficient and the difference between the 90th and 10th percentile of the consumption distribution. In Appendix E, we show that estimating a FAVAR leads to very similar conclusions about the transmission of surprise and news shocks to consumption inequality.

I KEPT IT REALLY CONCISE: SHOULD WE COMMENT ON THE MECHANISMS? I WOULD BE SILENT ON THEM IN THE EMPIRICAL SECTION..OTHERWISE WHAT'S THE POINT OF HAVING A MODEL?

3 Model

The model is based on the borrower-saver New Keynesian framework in Bilbiie et al. (2013), which has been used by Punzo and Rossi (2023) and Cantore and Meichtry (2024), among others. It consists of a household block with two types of households differing in their degree of impatience, a production block facing nominal frictions, government, and the central bank.[2] We analyse the impact of government spending shocks under two regimes: the monetary-led regime, wherein the government adheres to fiscal discipline and the central bank follows the Taylor Principle, and the fiscally-led regime, in which fiscal policy is exogenous, while the central bank accommodates it by passively setting interest rates. We show that the regimes yield contrasting distributional effects of government spending surprise and news shocks.

[2] In a standard TANK model, as in Bilbiie (2008), a fraction of households are hand-to-mouth by assumption; they simply consume their current income, displaying non-Ricardian behaviour. In the borrower–saver model of Bilbiie et al. (2013), all households are forward-looking optimisers, but some face a borrowing constraint; when it binds, they behave as if hand-to-mouth. If the borrowing constraint always binds and the share of borrowers is fixed, the borrower-saver model will look very similar to TANK in terms of consumption behaviour, as borrowers have a high MPC and consume close to current income. The key difference here, compared with standard TANK, is that constrained households still have balance sheets and debt positions, so inflation and interest rate changes affect them through real debt revaluation and redistribution. In the standard TANK, hand-to-mouth households typically have no assets or debt and no intertemporal margin, so those balance sheet and valuation channels are absent. Given this, we use the borrower-saver model for our main analysis. Our results are robust to the standard TANK model as well, and the results are presented in the Appendix.

3.1 Households

There is a continuum of households with a share λ of borrowers (B) who face an exogenous borrowing constraint, and $1-\lambda$ of savers (S) with unconstrained access to asset markets. Borrowers are assumed to be less patient than savers, implying that savers lend to the borrowers in equilibrium. Both households derive utility from consumption C and disutility from supplying labour L, and make an optimising labour supply decision, earning them wage income. Only the savers own shares in firms, hence receive profits. While both have access to financial markets, albeit with varying degrees, government bonds, however, are held only by the S agents.

3.1.1 Saver households

Saver households are Ricardian agents that discount future at the rate β^S and maximise utility

$$E_0 \sum_{t=0}^{\infty} \beta_t^S \left(\ln C_t^S - \frac{(L_t^S)^\eta}{\eta} \right), \tag{3.1}$$

where η is the inverse Frisch elasticity of labour supply, subject to the (real) budget constraint

$$C_t^S + B_t^S + \frac{B_t^G}{1-\lambda} = \frac{R_{t-1}B_{t-1}^S}{\pi_t} + \frac{R_{t-1}B_{t-1}^G}{\pi_t} + W_t L_t^S + \left(\frac{1-\tau^F}{1-\lambda}\right) F_t - \frac{1-\alpha}{1-\lambda} T_t. \tag{3.2}$$

They earn wage income W_t and profits F_t from owning monopolistically competitive firms, where τ^F is the parameter governing redistribution of profits. T_t are lump-sum taxes, R_t is the nominal interest income from lending B_t^S to borrower households and saving B_t^G in government bonds. Their optimality conditions are as follows:

$$\frac{1}{C_t^S} = \beta^S E_t \left[\frac{R_t}{\pi_{t+1} E_t C_{t+1}^S} \right], \tag{3.3}$$

$$W_t = (L_t^S)^{\eta-1} C_t^S \tag{3.4}$$

3.1.2 Borrower households

The preferences of borrowers are identical to the savers, except that they are relatively more impatient, as reflected by their discount factor β^B with $\beta^B < \beta^S$. They face a suitably defined exogenous borrowing constraint that binds each period. They maximise utility given by

$$E_0 \sum_{t=0}^{\infty} \beta_t^B \left(\ln C_t^B - \frac{(L_t^B)^\eta}{\eta} \right), \tag{3.5}$$

subject to the budget constraint

$$C_t^B + \frac{R_{t-1}B_{t-1}^B}{\pi_t} = B_t^B + W_t L_t^B + \left(\frac{\tau^F}{\lambda}\right) F_t - \frac{\alpha}{\lambda} T_t, \tag{3.6}$$

and the borrowing constraint

$$B_t^B \leq \bar{D}, \tag{3.7}$$

where $\bar{D} > 0$ is the exogenous borrowing limit. Due to their impatience, the equilibrium interest rate, which is determined by the more patient saver households' discount factor, induces them to borrow up to the limit. Their optimality conditions are as follows:

$$\frac{1}{C_t^B} = \beta^B E_t \left[\frac{R_t}{\pi_{t+1} E_t C_{t+1}^B}\right] + \lambda_t \tag{3.8}$$

$$W_t = (L_t^B)^{\eta-1} C_t^B \tag{3.9}$$

where λ_t is the Lagrange multiplier on the borrowing constraint (3.7) in the utility maximisation problem.

In the standard TANK model of Bilbiie (2008), $\bar{D}$ is zero, so B_t^B, λ_t, and β^B are zero as well. Hence, the only optimisation equation for this block, which would now represent hand-to-mouth agents H instead of borrowers with the same mass λ, would be the budget constraint and labour supply equation:

$$C_t^H = W_t L_t^H + \left(\frac{\tau^F}{\lambda}\right) F_t - \frac{\alpha}{\lambda} T_t, \tag{3.10}$$

$$W_t = (L_t^H)^{\eta-1} C_t^H \tag{3.11}$$

3.2 Firms

The production block of the model is relatively standard. The model incorporates two types of firms, monopolistically competitive producers of differentiated intermediate goods and perfectly competitive final goods producers. Intermediate goods producers utilise labour using CRS technology as per the production function, $Y_t = L_t$, where L_t is the aggregate labour supplied by the two types of households.

They face price rigidity as in Calvo (1983) with probability $(1-\theta)$ of resetting prices. Each period, firms choose the price to maximize the expected present discounted value of profits

discounted by the Savers' stochastic discount factor, $\Lambda_{t,k} = \beta^S \frac{C_t^S}{C_{t+k}^S}$. In equilibrium, all firms behave identically, and the optimality condition for price-setting is given by

$$\sum_{k=0}^{\infty} (\beta_0 \theta)^k E_t \left[\Lambda_{t,k} \left(\frac{P_t^*(z)}{P_{t+k}} - \frac{\varepsilon}{\varepsilon - 1} X_{t+k} \right) Y_{t+k}^*(z) \right] = 0, \tag{3.12}$$

where $X = \frac{\varepsilon}{\varepsilon - 1}$ and ε is elasticity of substitution between varieties of intermediate goods.

The intermediate goods producers maximise their profits $F_t = \left(1 - \frac{1}{X_t}\right)$ to be paid out to the Ricardian households. The final goods sector is competitive and aggregates these differentiated intermediate goods according to a CES technology standard in New Keynesian frameworks.

3.3 Fiscal Policy

The government spends G_t, levies lump-sum taxes on households T_t, and issues one-period bonds B_t^G, which are held by Ricardian households only. The fiscal block comprises the intertemporal budget constraint, a tax rule, and an AR(1) process for government spending, given by

$$B_t^G = \frac{R_{t-1}}{\pi_t} B_{t-1}^G + G_t - T_t, \tag{3.13}$$

$$\frac{T_t}{T} = \left(\frac{T_{t-1}}{T} \right)^{\rho_T} \left(\frac{B_{t-1}^G}{B^G} \right)^{(1-\rho_T)\phi_{TB}} \left(\frac{G_t}{G} \right)^{(1-\rho_T)\phi_{TB}} \epsilon_T, \tag{3.14}$$

$$\frac{G_t}{G} = \left(\frac{G_{t-1}}{G} \right)^{\rho_G} \epsilon_{G,t-j} \tag{3.15}$$

where variables without time subscripts denote steady-state values, $\epsilon_{G,t-j}$ is an iid government spending shock with $j \geq 0$, where $j = 0$ denotes an unanticipated (surprise) shock and $j > 0$ denotes an anticipated (news) shock to government spending, and ϵ_T is the iid tax shock.

3.4 Monetary Policy

The central bank follows a Taylor-type rule setting nominal interest rates in response to inflation and the output gap as follows

$$\frac{R_t}{R} = \left(\frac{R_{t-1}}{R} \right)^{\rho_R} \left(\frac{\pi_t}{\pi} \right)^{(1-\rho_R)\phi_\pi} \left(\frac{y_t}{y} \right)^{(1-\rho_R)\phi_y} \epsilon_{R,t}, \tag{3.16}$$

where ρ_R is the interest rate smoothing parameter, ϕ_π and ϕ_y are the responsiveness coefficients with $\rho_R, \phi_\pi, \phi_y \geq 0$, and ϵ_R is an iid monetary policy shock.

3.5 Market clearing and equilibrium

In equilibrium of this economy, all agents, except monopolistic price-setting firms, take prices as well as the evolution of exogenous processes as given. A rational expectations equilibrium is then a sequence of processes for all prices and quantities such that the optimality conditions hold for all agents and all markets clear at any given time t. Aggregate output is the sum of private spending and public spending, $(1-\lambda)C_t^S + \lambda C_t^B + G_t = Y_t$, and labour and bonds markets clear such that $(1-\lambda)L_t^S + \lambda L_t^B = L_t$ and $(1-\lambda)B_t^S = \lambda B_t^B$.

We solve the model around a zero net inflation steady state. In what follows, lowercase letters denote the log deviation of a variable from its steady state. The log-linearised model is presented in Table 1.

Table 1: Summary of the log-linearised model

Description	Equation
Budget constraint, B	$\frac{C^B}{Y}c_t^B + \frac{\bar{D}}{Y}(r_{t-1} - \pi_t) = \frac{WL}{Y}(w_t + l_t^B) + \frac{F}{Y}\frac{\tau^D}{\lambda}f_t - \frac{T}{Y}t_t$
OR Budget constraint, H	$\frac{C^H}{Y}c_t^H = \frac{WL}{Y}(w_t + l_t^H) + \frac{F}{Y}\frac{\tau^D}{\lambda}f_t - \frac{T}{Y}t_t$
Euler equation, S	$c_t^S = E_t c_{t+1}^S - \sigma r_t$
Labour supply equation	$\eta l_t = w_t - c_t$
Aggregate Output	$y_t = \frac{C}{Y}c_t + \frac{G}{Y}g_t$
Production function	$y_t = l_t$
Real marginal cost	$mc_t = w_t$
Profits	$d_t = -w_t$
Phillips curve	$\pi_t = \beta E_t \pi_{t+1} + \kappa y_t$
Government budget constraint	$b_{t+1}^G = \frac{1}{\beta^S}(b_t^G + r_{t-1}) + \frac{G}{B}g_t - \frac{T}{B}t_t$
Total borrowing/lending	$(1-\lambda)b_t^S = \lambda b_t^B$
Tax rule	$t_t = \rho_T t_{t-1} + (1-\rho_T)(\phi_{TB}b_t + \phi_{TG}g_t) - \varepsilon_t^T$
Government spending shock	$g_t = \rho_G g_{t-1} + \varepsilon_{G,t-j}, j \geq 0$
Taylor rule	$r_t = \rho_r r_{t-1} + (1-\rho_r)(\phi_\pi \pi_t + \phi_y y_t) + \varepsilon_t^M$

3.6 Response to anticipated and unanticipated government spending shocks

3.6.1 An analytical illustration - SB TANK

In this section, we provide a simple analytical illustration with the borrower-saver TANK framework with S and B agents to understand the impact of government spending surprise

and news shocks on consumption inequality. Define consumption inequality between S and B, as

$$\gamma_t^{SB} \equiv c_t^S - c_t^B = \frac{1-\chi\Gamma}{1-\lambda\delta}c_t - \frac{z\chi}{1-\lambda\delta}g_t + \frac{\alpha}{\lambda}\frac{z}{1-\lambda\delta}t_t, \tag{3.17}$$

where $\chi = \left(\left(1 - \frac{\tau^D}{\lambda}\right)\eta + 1\right)$, $\Gamma = \frac{1+\eta}{1+\eta\delta}$, and $z = \frac{\eta}{1+\eta\delta}$.

The parameter χ governing the profit redistribution from savers to borrower households has received special attention in the various works of Bilbiie (2020). Combining the saver's Euler and labour supply condition with market-clearing conditions, we get the following (log-linearised) IS equation for the economy

$$c_t = E_t c_{t+1} + \frac{\lambda\delta\chi z}{1-\lambda\delta\chi\Gamma}(E_t g_{t+1} - g_t) + \frac{\lambda z}{1-\lambda\delta\chi\Gamma}\frac{\alpha}{\lambda}(E_t t_{t+1} - t_t) - \sigma\frac{1-\lambda\delta}{1-\lambda\delta\chi\Gamma}r_t. \tag{3.18}$$

Assume the government has a balanced budget such that its budget constraint, equation 3.13, simplifies to $g_t = t_t$, and further we have uniform taxation such that $\alpha = \lambda$, we get the simplified forwarded (to finite N) aggregate IS as follows:

$$c_t = E_t c_{t+N} - \frac{\lambda z}{1-\lambda\delta\chi\Gamma}(\delta\chi - 1)\sum_{j=0}^{N-1} E_t\left(g_{t+j+1} - g_{t+j}\right) - \sigma\frac{1-\lambda\delta}{1-\lambda\delta\chi\Gamma}\sum_{j=0}^{N-1} E_t r_{t+j}. \tag{3.19}$$

Notice $\sum_{j=0}^{N-1} E_t\left(g_{t+j+1} - g_{t+j}\right) = (E_t g_{t+N} - g_t)$ in equation 3.29. Combining equations 3.26 and 3.29 for the balanced budget case, we can evaluate the impact of government spending news and surprise shocks on consumption inequality using the following

$$\begin{aligned}\gamma_t^{SB} = {} & \frac{1-\chi\Gamma}{1-\lambda\delta}\left(E_t c_{t+N} - \frac{\lambda z}{1-\lambda\delta\chi\Gamma}(\delta\chi - 1)(E_t g_{t+N} - g_t) - \sigma\frac{1-\lambda\delta}{1-\lambda\delta\chi\Gamma}\sum_{j=0}^{N-1} E_t r_{t+j}\right) \\ & - \frac{z}{1-\lambda\delta}(\chi - 1)g_t. \end{aligned} \tag{3.20}$$

In the case of countercyclical inequality i.e. $\chi > 1$, and further assuming $\lambda\chi < 1$ as in Bilbiie (2020), ceteris paribus, on impact, a government spending news shock increases consumption inequality, while a government spending surprise shock decreases inequality, evaluated as follows

$$\frac{\partial\gamma_0^{SB}}{\partial g_N} = -\frac{(1-\chi\Gamma)}{1-\lambda\delta}\frac{\lambda z}{1-\lambda\delta\chi\Gamma}(\delta\chi - 1) \text{ and } \frac{\partial\gamma_0^{SB}}{\partial g_0} = \frac{(1-\chi\Gamma)}{1-\lambda\delta}\frac{\lambda z}{1-\lambda\delta\chi\Gamma}(\delta\chi - 1) - \frac{z}{1-\lambda\delta}(\chi - 1). \tag{3.21}$$

The general equilibrium effect of government spending news and surprise shocks is described in the following propositions.

Proposition 1: Consider a pure government spending news shock at time $t = 0$, defined as a marginal increase in expected future spending $E_0 g_{t+N}$, with no contemporaneous change in g_0. Using equations 3.26 and 3.29, the equilibrium announcement effect on inequality is

$$\frac{d\gamma_0^{SB}}{dg_N} = \frac{1-\chi\Gamma}{1-\lambda\delta}\frac{dc_0}{dg_N} \tag{3.22}$$

Given this, when $\frac{dc_0}{dg_N} > 0$, then a government spending news shock decreases inequality with $\chi > 1$ and increases inequality when $\chi < 1$.

Proposition 2: Consider a pure government spending surprise shock at time $t = 0$, defined as a realised increase in government spending $E_0 g_0$. Using equations 3.26 and 3.29, the equilibrium announcement effect on inequality is

$$\frac{d\gamma_0^{SB}}{dg_0} = \frac{1-\chi\Gamma}{1-\lambda\delta}\frac{dc_0}{dg_0} + \frac{z}{1-\lambda\delta}(\chi - 1). \tag{3.23}$$

Given this, when $\frac{dc_0}{dg_0} > 0$, then a government spending surprise shock decreases inequality with $\chi > 1$ and increases inequality when $\chi < 1$.

The change in consumption at $t = 0$, for both government spending news and surprise shocks, needs further examination. Evaluating this using equation 3.28, we show that this is a composite of three effects

$$\frac{dc_0}{dg_N} = \underbrace{\frac{dE_0 c_N}{dg_N}}_{\Psi_C} - \underbrace{\frac{\lambda z}{1-\lambda\delta\chi\Gamma}(\delta\chi - 1)}_{\Psi_G} - \underbrace{\sigma\frac{1-\lambda\delta}{1-\lambda\delta\chi\Gamma}\frac{d\sum_{j=0}^{N-1} E_0 r_j}{dg_N}}_{\Psi_R}, \tag{3.24}$$

where Ψ_C measures how much expected future consumption at N affects consumption today when the announcement is made, and Ψ_G is the direct marginal effect of government spending increase announced today, but to be implemented N periods ahead. Ψ_R is the change in the sum of expected real rates over the next N periods, which alters the relative price of consumption today versus tomorrow when expected real rates are revised after the announcement.

Given this, anticipated future government spending affects current consumption through two channels: first, directly through the change in expected future consumption as well as anticipation of future government spending, and second, the standard intertemporal substitution channel through interest rates. Relative to a standard representative agent framework, both these effects are scaled by redistribution governed by $\lambda\chi$. However, from 3.28, $\Psi_C = \Psi_G$ in our simple model, which implies that the impact of the news shock is purely driven by the change

in interest rates. With $\lambda\chi < 1$, $\frac{dc_0}{dg_N} > 0$ when the real rate is reduced in response to a news shock.

Next, we examine the consumption impact of a surprise shock as follows:

$$\frac{dc_0}{dg_0} = \underbrace{\frac{dE_0 c_N}{dg_0}}_{\Psi_{CC}} + \underbrace{\frac{\lambda z}{1-\lambda\delta\chi\Gamma}(\delta\chi - 1)}_{\Psi_{GG}} - \underbrace{\sigma\frac{1-\lambda\delta}{1-\lambda\delta\chi\Gamma}\frac{d\sum_{j=0}^{N-1} E_0 r_j}{dg_0}}_{\Psi_{RR}}, \tag{3.25}$$

In the above expression, Ψ_{CC} is the change in future consumption at period N, Ψ_{GG} is the contemporaneous change in consumption to a change in government spending or the contemporaneous fiscal multiplier, and Ψ_{RR} is the associated interest rate response. Absent persistence of government spending it is fair to assume $\Psi_{CC} = 0$, $\Psi_{GG} > 0$ with $\chi > 1$ and $\lambda\chi < 1$. The sign of Ψ_{RR} is defined by the real rate response, as its coefficient is negative. Hence, the overall magnitude of $\frac{dc_0}{dg_0}$ is governed by the fiscal multiplier on consumption net of the interest rate response.

3.6.2 An analytical illustration - ORIGINAL TANK

In this section, we provide a simple analytical illustration with the standard TANK model with S and H agents to understand the impact of government spending surprise and news shocks on consumption inequality. Define consumption inequality between S and H, as

$$\gamma_t^{SH} \equiv c_t^S - c_t^H = \frac{1-\chi}{1-\lambda}c_t - z\chi q g_t + \frac{\alpha}{\lambda} z q t_t, \tag{3.26}$$

where $\chi = \left(\left(1 - \frac{\tau^D}{\lambda}\right)\eta + 1\right)$, $q = \left(\frac{\lambda}{1-\lambda} + 1\right)$ and $z = \frac{\eta}{1+\eta}$.

The parameter χ governing the profit redistribution from savers to hand-to-mouth households has received special attention in the various works of Bilbiie (2020). Combining the saver's Euler and labour supply condition with market-clearing conditions, we get the following (log-linearised) IS equation for the economy

$$c_t = E_t c_{t+1} - \frac{\lambda\chi z}{1-\lambda\chi}(g_t - E_t g_{t+1}) - \frac{\lambda z}{1-\lambda\chi}\frac{\alpha}{\lambda}(t_t - E_t t_{t+1}) - \sigma\frac{1-\lambda}{1-\lambda\chi} r_t. \tag{3.27}$$

from bilbiie, chapter 3, page 16

$$c_t = E_t c_{t+1} + \frac{\lambda\chi z}{1-\lambda\chi}(g_t - E_t g_{t+1}) - \frac{\lambda z}{1-\lambda\chi}\frac{\alpha}{\lambda}(t_t - E_t t_{t+1}) - \sigma\frac{1-\lambda}{1-\lambda\chi} r_t. \tag{3.28}$$

Assume the government has a balanced budget such that its budget constraint, equation 3.13, simplifies to $g_t = t_t$, and further we have uniform taxation such that $\alpha = \lambda$, we get the simplified

forwarded (to finite N) aggregate IS as follows:

$$c_t = E_t c_{t+N} - \frac{\lambda z}{1-\lambda\chi}(\chi - 1)\sum_{j=0}^{N-1} E_t\left(g_{t+j+1} - g_{t+j}\right) - \sigma\frac{1-\lambda}{1-\lambda\chi}\sum_{j=0}^{N-1} E_t r_{t+j}. \tag{3.29}$$

Notice $\sum_{j=0}^{N-1} E_t\left(g_{t+j+1} - g_{t+j}\right) = (E_t g_{t+N} - g_t)$ in equation 3.29. Combining equations 3.26 and 3.29 for the balanced budget case, we can evaluate the impact of government spending news and surprise shocks on consumption inequality using the following

$$\gamma_t = \frac{1-\chi}{1-\lambda}\left(E_t c_{t+N} - \frac{\lambda z}{1-\lambda\chi}(\chi-1)(E_t g_{t+N} - g_t) - \sigma\frac{1-\lambda}{1-\lambda\chi}\sum_{j=0}^{N-1} E_t r_{t+j}\right) - z(\chi-1)qg_t. \tag{3.30}$$

In the case of countercyclical inequality i.e. $\chi > 1$, and further assuming $\lambda\chi < 1$ as in Bilbiie (2020), ceteris paribus, on impact, a government spending news shock increases consumption inequality, while a government spending surprise shock decreases inequality, evaluated as follows

$$\frac{\partial\gamma_0}{\partial g_N} = \frac{(\chi-1)^2}{1-\lambda}\frac{\lambda z}{1-\lambda\chi} \quad \text{and} \quad \frac{\partial\gamma_0}{\partial g_0} = -\frac{(\chi-1)^2}{1-\lambda}\frac{\lambda z}{1-\lambda\chi} - z(\chi-1)qg_t. \tag{3.31}$$

The general equilibium effect of government spending news and surprise shocks is described in the following propositions.

Proposition 1: Consider a pure government spending news shock at time $t = 0$, defined as a marginal increase in expected future spending $E_0 g_{t+N}$, with no contemporaneous change in g_0. Using equations 3.26 and 3.29, the equilibrium announcement effect on inequality is

$$\frac{d\gamma_0}{dg_N} = \frac{1-\chi}{1-\lambda}\frac{dc_0}{dg_N} \tag{3.32}$$

Given this, when $\frac{dc_0}{dg_N} > 0$, then a government spending news shock decreases inequality with $\chi > 1$ and increases inequality when $\chi < 1$.

Proposition 2: Consider a pure government spending surprise shock at time $t = 0$, defined as a realised increase in government spending $E_0 g_0$. Using equations 3.26 and 3.29, the equilibrium announcement effect on inequality is

$$\frac{d\gamma_0}{dg_0} = \frac{1-\chi}{1-\lambda}\frac{dc_0}{dg_0} + zq(1-\chi) \tag{3.33}$$

Given this, when $\frac{dc_0}{dg_0} > 0$, then a government spending surprise shock decreases inequality with $\chi > 1$ and increases inequality when $\chi < 1$.

The change in consumption at $t = 0$, for both government spending news and surprise shocks, needs further examination. Evaluating this using equation 3.28, we show that this is a composite of three effects

$$\frac{dc_0}{dg_N} = \underbrace{\frac{dE_0 c_N}{dg_N}}_{\Psi_C} - \underbrace{\frac{\lambda z}{1-\lambda\chi}(\chi - 1)}_{\Psi_G} - \underbrace{\sigma \frac{1-\lambda}{1-\lambda\chi} \frac{d\sum_{j=0}^{N-1} E_0 r_j}{dg_N}}_{\Psi_R}, \tag{3.34}$$

where Ψ_C measures how much expected future consumption at N affects consumption today when the announcement is made, and Ψ_G is the direct marginal effect of government spending increase announced today, but to be implemented N periods ahead. Ψ_R is the change in the sum of expected real rates over the next N periods, which alters the relative price of consumption today versus tomorrow when expected real rates are revised after the announcement.

Given this, anticipated future government spending affects current consumption through two channels: first, directly through the change in expected future consumption as well as anticipation of future government spending, and second, the standard intertemporal substitution channel through interest rates. Relative to a standard representative agent framework, both these effects are scaled by redistribution governed by $\lambda\chi$. However, from 3.28, $\Psi_C = \Psi_G$ in our simple model, which implies that the impact of the news shock is purely driven by the change in interest rates. With $\lambda\chi < 1$, $\frac{dc_0}{dg_N} > 0$ when the real rate is reduced in response to a news shock.

XXXXXXX

Next, we examine the consumption impact of a surprise shock as follows:

$$\frac{dc_0}{dg_0} = \underbrace{\frac{dE_0 c_N}{dg_0}}_{\Psi_{CC}} + \underbrace{\frac{\lambda z}{1-\lambda\chi}(\chi - 1)}_{\Psi_{GG}} - \underbrace{\sigma \frac{1-\lambda}{1-\lambda\chi} \frac{d\sum_{j=0}^{N-1} E_0 r_j}{dg_0}}_{\Psi_{RR}}, \tag{3.35}$$

In the above expression, Ψ_{CC} is the change in future consumption at period N, Ψ_{GG} is the contemporaneous change in consumption to a change in government spending or the contemporaneous fiscal multiplier, and Ψ_{RR} is the associated interest rate response. Absent persistence of government spending it is fair to assume $\Psi_{CC} = 0$, $\Psi_{GG} > 0$ with $\chi > 1$ and $\lambda\chi < 1$. The sign of Ψ_{RR} is defined by the real rate response, as its coefficient is negative. Hence, the overall magnitude of $\frac{dc_0}{dg_0}$ is governed by the fiscal multiplier on consumption net of the interest rate response.

3.6.3 Numerical Results

In this section, we analyse the impact of government spending surprise and news shocks under two alternative policy regimes. Table 2 presents the calibration used in the model. We assume log utility and unitary Frisch elasticity of labour supply. The steady-state share of government

spending in output (20%) and the annual debt-to-output ratio (57%) are set to match the average for the US economy from 1984 to 2018. In all simulations, in the monetary-led regime, the Taylor rule coefficient on inflation, ϕ_π is set as 1.5 while it is 0.9 in the fiscally-led regime. Tax rule coefficients on government spending and borrowing are set to be $\phi_{TG} = 0.1$ and $\phi_{TB} = 0.33$, respectively in the monetary-led regime as in Galí et al. (2007a), while they are $\phi_{TG} = 0 = \phi_{TB}$ in the fiscally-led regime.

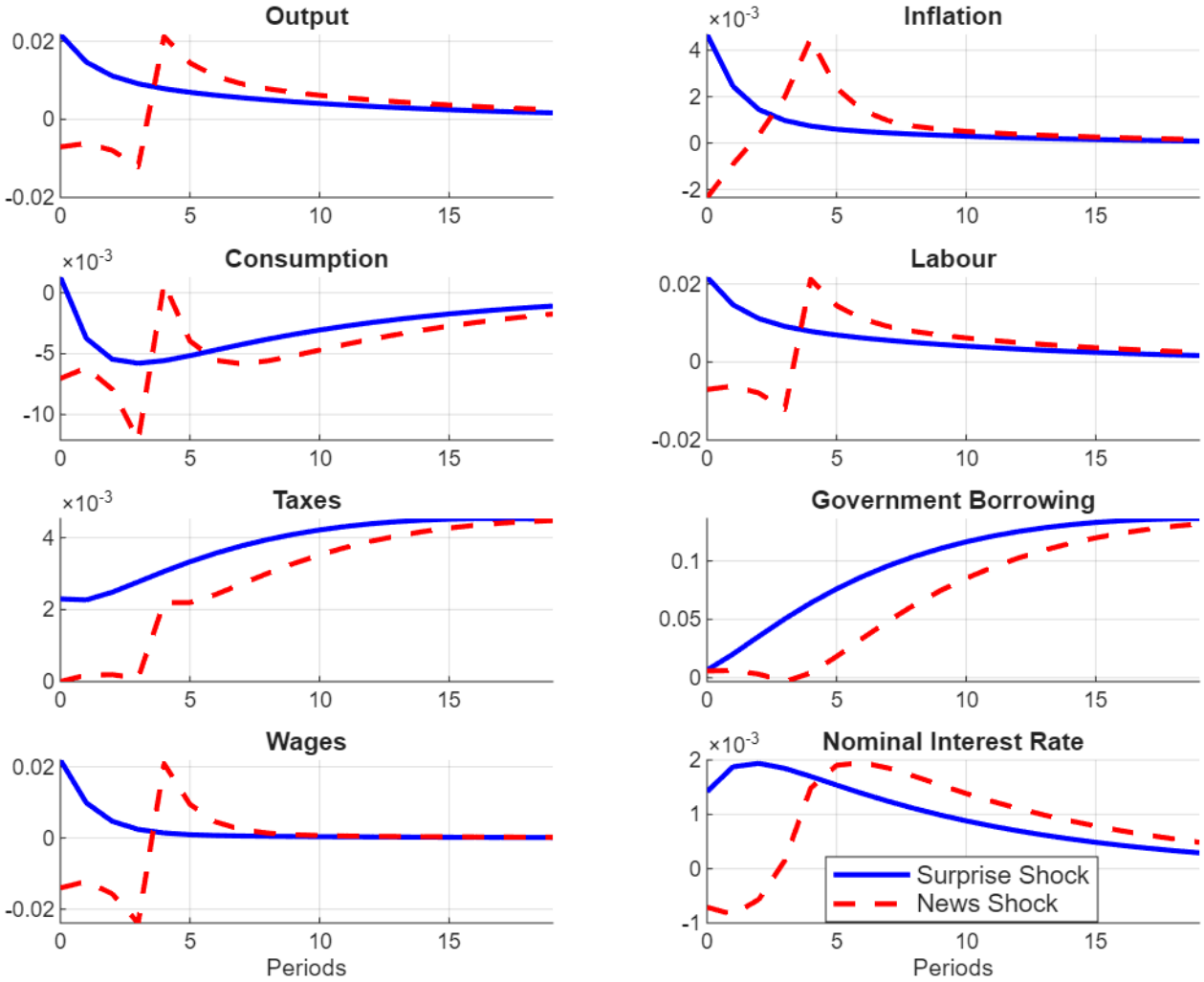
Table 2: Calibration

Parameter	**Interpretation**	**Value**
β^S	Saver discount factor	0.99
β^B	Borrower HtM discount factor	0.95
λ	Population share of the borrower HHs	0.35
X	Steady-state markup	1.05
θ	Calvo parameter	0.75
G/Y	Steady-state nominal government spending-to-GDP ratio	0.2
B/Y	Steady-state nominal debt-to-GDP ratio	0.57
ρ_R	Interest rate persistence/smoothing parameter	0.8
ρ_T	Tax rule persistence/smoothing parameter	0
ρ_G	Government spending persistence/smoothing parameter	0.9
σ_G	Government spending shock standard deviation	0.01

In the monetary-led regime (regime M), the central bank follows the Taylor Principle to raise the real rate in response to an increase in inflation, while the government is committed to satisfying its present value budget constraint at all times. In the fiscally-led regime (regime F), the government spends and taxes exogenously while the central bank passively sets interest rates to accommodate fiscal needs. The behaviour of the model under these two shocks is contingent on the precise policy mix in place. Broadly, we find that in the monetary-led regime, the surprise and news shocks have contrasting distributional implications. Surprise shocks reduce consumption inequality, and news shocks increase it. On the other hand, in the fiscally-led regime, the two shocks are qualitatively indistinguishable. This is described in detail as follows.

In regime M, as shown in Figure 3, a positive government spending surprise shock increases demand in the economy. Given sticky prices, this leads to an increase in inflation, wages and hours worked; the overheating in the economy calls for higher interest rates by the central bank. There is also an increase in tax liabilities and government borrowing. The savers, in line with Ricardian behaviour, reduce their consumption and increase savings in government bonds. Borrowers, on the other hand, consume more as the increase in wage income outpaces the increase in interest expenses. As a result, consumption inequality decreases, as shown in Figure 4.

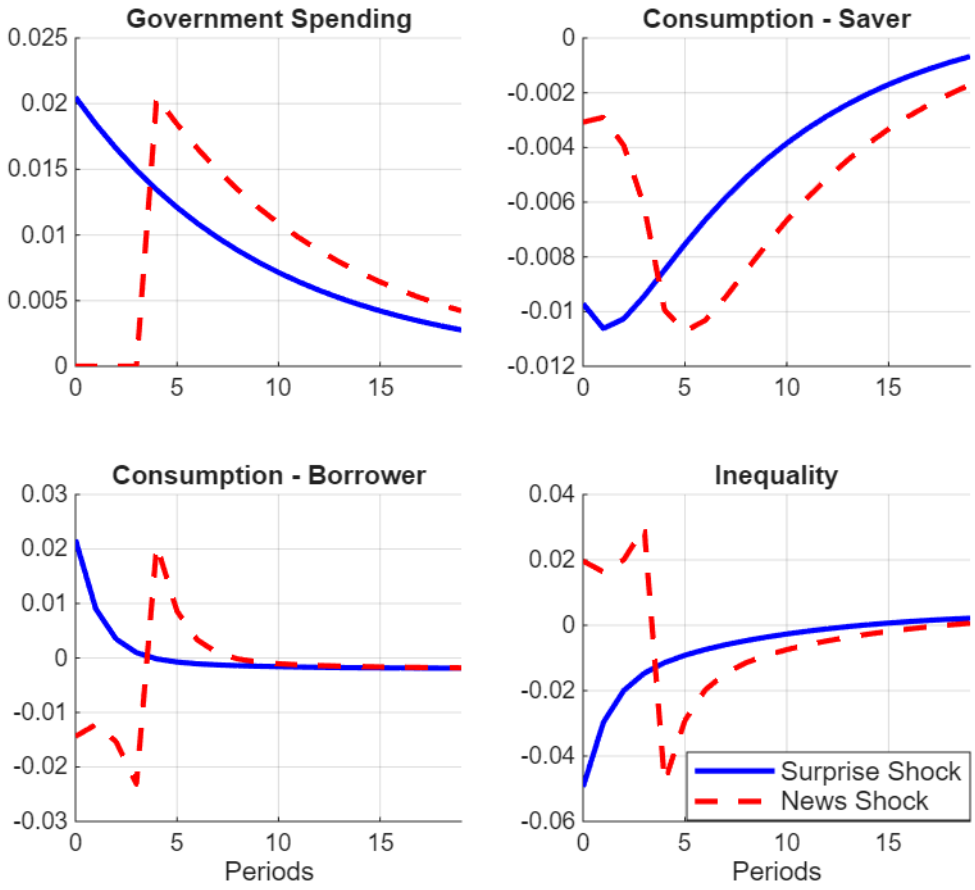
Figure 3: Regime M: Responses to a surprise and news government spending shock



Notes: The figure shows the impulse responses of selected variables to an anticipated government spending shock (news shock) vs. an unanticipated government spending shock (surprise shock) in the monetary regime. All series are in per cent deviations from their steady state except for the interest rate, which is in absolute deviation from the steady state.

A positive news shock in regime M, as shown in Figure 3, is contractionary as consumption, output, and hours worked contract in anticipation of future tax liabilities and government borrowing. This echoes the findings in Ascari et al. (2023). The contraction leads to a fall in inflation and warrants a lower interest rate as per the Taylor rule. Consumption of both savers and borrowers falls, but the latter are disproportionately worse off due to the fall in wages. The savers are relatively insulated as they also experience an increase in profits. As a result, upon announcement, consumption inequality between savers and borrowers rises, see 4.

Figure 4: Regime M: Effect on consumption inequality



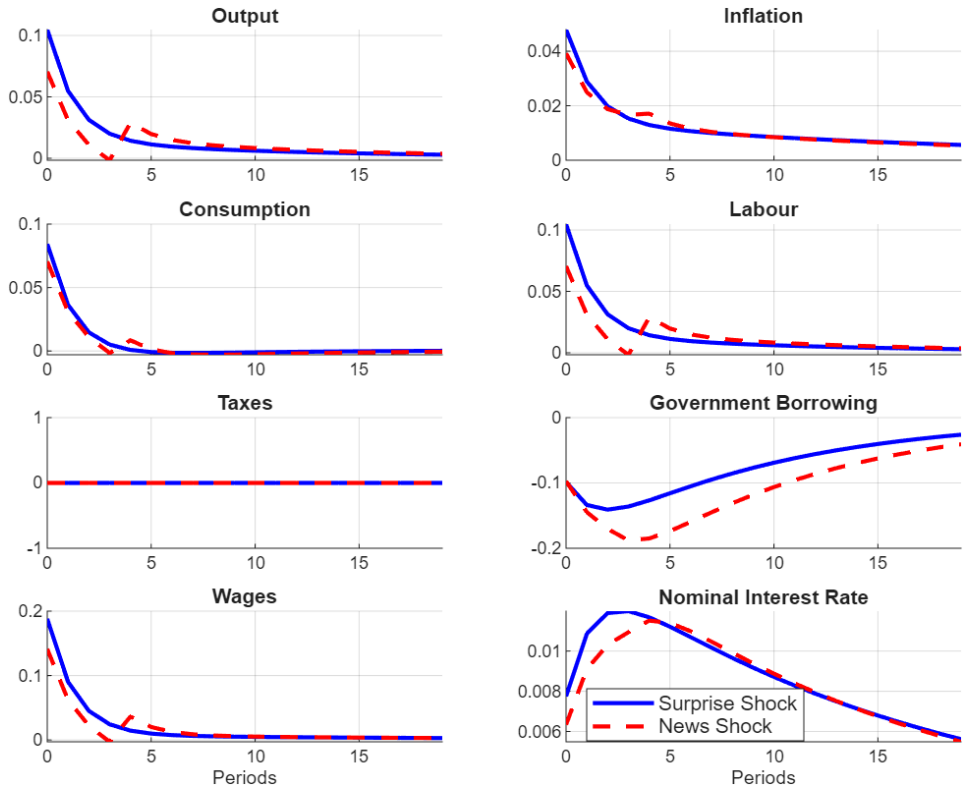
Notes: The figure shows the impulse responses of selected variables to an anticipated government spending shock (news shock) vs. an unanticipated government spending shock (surprise shock) in the monetary regime.

It is important to note that consumption inequality starts to fall once the actual government spending increase is implemented, showing a similar trajectory as the surprise shock.

In regime F, as shown in Figure 5, the model behaves very similarly qualitatively in both surprise and news shocks. The savers consider both the actual and anticipated government spending increases as a wealth gain due to the active fiscal policy stance, where spending increases are not matched by an increase in taxes or borrowing. Consumption, output, hours worked, wages, and inflation increase. The increase in the interest rate is also less than the increase in inflation, so the real rate actually goes down. This boosts consumption of both agents, and consumption inequality decreases for both surprise and news shocks. This is shown in Figure 6.

These results underscore that the interaction between monetary and fiscal policy not only shapes aggregate dynamics, as is well known since Leeper (1991), but also determines the

Figure 5: Regime F: Responses to a surprise and news government spending shock



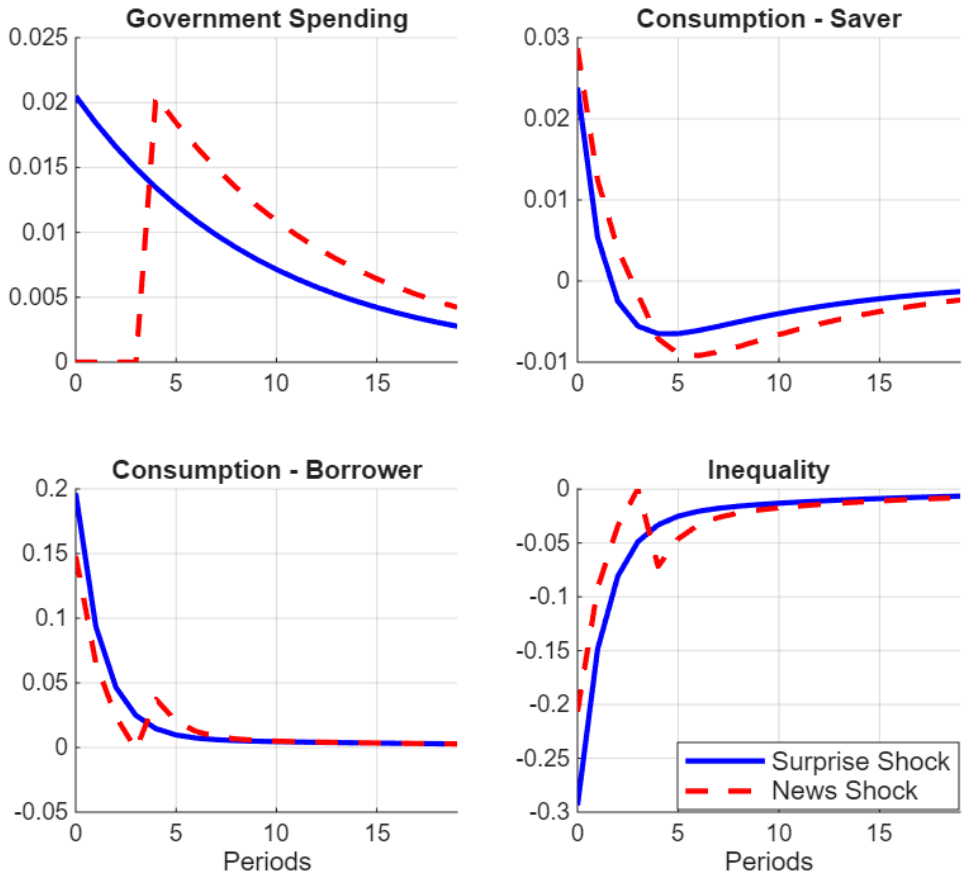
Notes: The figure shows the impulse responses of selected variables to an anticipated government spending shock (news shock) vs. an unanticipated government spending shock (surprise shock) in the fiscal regime. All series are in per cent deviations from their steady state except for the interest rate, which is in absolute deviation from the steady state.

distributional consequences of fiscal shocks. The policy regime governs whether anticipation effects amplify or dampen inequality, highlighting the importance of coordinated policy design in managing both macroeconomic stability and distributional outcomes.

4 Conclusion (F+R+V)

This paper examines the effect of government spending shocks on consumption inequality, distinguishing between unanticipated (surprise) and anticipated (news) shocks. Using the identification strategy of Forni and Gambetti (2016), our results reveal starkly different distributional consequences of the two shocks. Consistent with previous research, positive surprise shocks reduce consumption inequality. By contrast, we document a novel finding that news shocks increase consumption inequality. We show that this finding could be rationalised in a borrower–saver TANK model in which the effects of fiscal shocks depend crucially on the prevailing monetary–fiscal

Figure 6: Regime F: Effect on consumption inequality



Notes: The figure shows the impulse responses of selected variables to an anticipated government spending shock (news shock) vs. an unanticipated government spending shock (surprise shock) in the fiscal regime.

policy regime.

In the monetary-led regime, where the central bank follows the Taylor principle and fiscal policy satisfies its intertemporal budget constraint, surprise and news shocks have contrasting effects. A surprise spending increase stimulates demand and raises wages, benefiting borrowers relatively more and reducing consumption inequality. In contrast, a news shock is initially contractionary, as households anticipate future fiscal adjustments; wages fall and borrowers are disproportionately worse-off, leading to a temporary rise in inequality until the spending materialises. In the fiscally-led regime, where fiscal policy is active and monetary policy accommodates it, surprise and news shocks produce similar outcomes. Both types of shocks are expansionary, as higher government spending is perceived as a net wealth gain. Real interest rates decline, consumption rises for both savers and borrowers, and consumption inequality falls in both cases.

Overall, our findings highlight that the distributional consequences of fiscal policy depends on both their timing and on the prevailing monetary–fiscal policy mix. Accounting for anticipation effects and policy regimes is therefore essential for understanding how fiscal expansions shape inequality and better coordinating monetary and fiscal policies.

This paper examines the effect of government spending shocks on US consumption inequality. We treat unanticipated (surprise) and news (anticipated) shocks to government spending as distinct, and separately identify them using the methodology proposed by Forni and Gambetti (2016).

We show that the macroeconomic and distributional effects of government spending shocks depend critically on the policy regime and on whether shocks are anticipated or unanticipated. In the monetary-led regime, where the central bank follows the Taylor principle and fiscal policy is Ricardian, surprise and news shocks have contrasting effects. A surprise spending increase stimulates demand and inflation, prompting monetary tightening; borrowers benefit from higher wages while savers cut consumption, leading to a decline in consumption inequality. By contrast, a news shock is initially contractionary, as Ricardian households anticipate future fiscal adjustments. Output and wages fall, borrowers are hit harder than savers, and consumption inequality rises until the spending is implemented.

In the fiscally-led regime, where fiscal policy is active and monetary policy accommodates it, surprise and news shocks produce similar outcomes. Higher government spending is perceived as a net wealth gain, real rates fall, and consumption increases for both savers and borrowers. As a result, consumption inequality declines in response to both types of shocks.

Overall, the findings underscore that the interaction between monetary and fiscal policy not only shapes aggregate dynamics but also determines the distributional consequences of fiscal shocks. The policy regime governs whether anticipation effects amplify or dampen inequality, highlighting the importance of coordinated policy design in managing both macroeconomic stability and distributional outcomes.

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A Data appendix

- Federal government consumption expenditures and gross investment (BEA: B823RG)
- Gross domestic product (BEA: A191RC)
- Federal budget surplus: receipts (BEA: W018RC1) minus spending (BEA: W019RC1)
- 10-year Treasury constant maturity rate (FRED: DGS10)
- Federal funds rate (FRED: DFF).
- Real exchange rate (FRED: TWEXM)
- Corporate profits (after tax) (FRED: CP)
- Consumer confidence (FRED: CSCICP03USM665S)
- GDP deflator (BEA: A191RI)

B Informational sufficiency

Following Forni and Gambetti (2014), we document the informational sufficiency of our baseline VAR model by regressing the identified surprise and news shocks to government spending on the first seven principal components from the FRED-QD (McCracken and Ng, 2021). This is a large quarterly database, publicly available on the Federal Reserve Bank of St. Louis's website, which collects 246 U.S. macroeconomic and financial variables. Table B.3 shows p-values from F-tests assessing whether each shock is orthogonal to lags (up to 4) of the principal components. None of the common factors is found to have predictive power for either of the two shocks.

Surprise shock

PC	Lag 1	Lag 1:2	Lag 1:3	Lag 1:4
1	0.59	0.66	0.83	0.92
2	0.18	0.35	0.48	0.65
3	0.86	0.90	0.89	0.96
4	0.67	0.25	0.28	0.37
5	0.50	0.76	0.44	0.17
6	0.40	0.70	0.11	0.19
7	0.79	0.63	0.79	0.83

News shock

PC	Lag 1	Lag 1:2	Lag 1:3	Lag 1:4
1	0.82	0.71	0.86	0.94
2	0.26	0.51	0.66	0.33
3	0.65	0.53	0.73	0.62
4	0.85	0.86	0.38	0.47
5	0.60	0.64	0.76	0.72
6	0.79	0.64	0.83	0.92
7	0.51	0.68	0.72	0.49

Table B.3: Testing for informational sufficiency (medium-scale VAR), * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

C Structural shocks series

C.1 Surprise shock

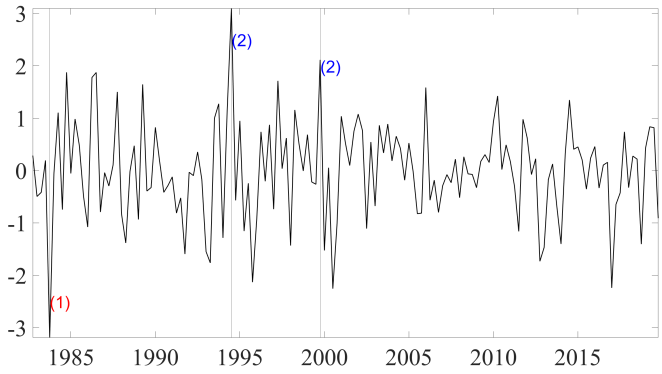

Figure C.7: Median surprise shock.

- Negative: (1) Reagan's nondefense spending cuts (1983Q4).
- Positive: (2) Military procurement upticks (1994Q3, 1999Q4).

C.2 News shock

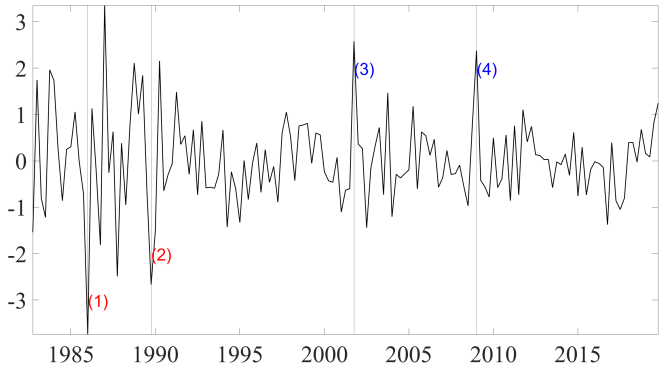

Figure C.8: Median news shock.

- Negative: (1) Emergency Deficit Control Act (1986Q1); (2) Berlin Wall fall (1989Q4).
- Positive: (3) Afghanistan War (2001Q4); (4) Obama Fiscal Stimulus (2009Q1).

D Robustness checks

D.1 $n_t(1,4)$

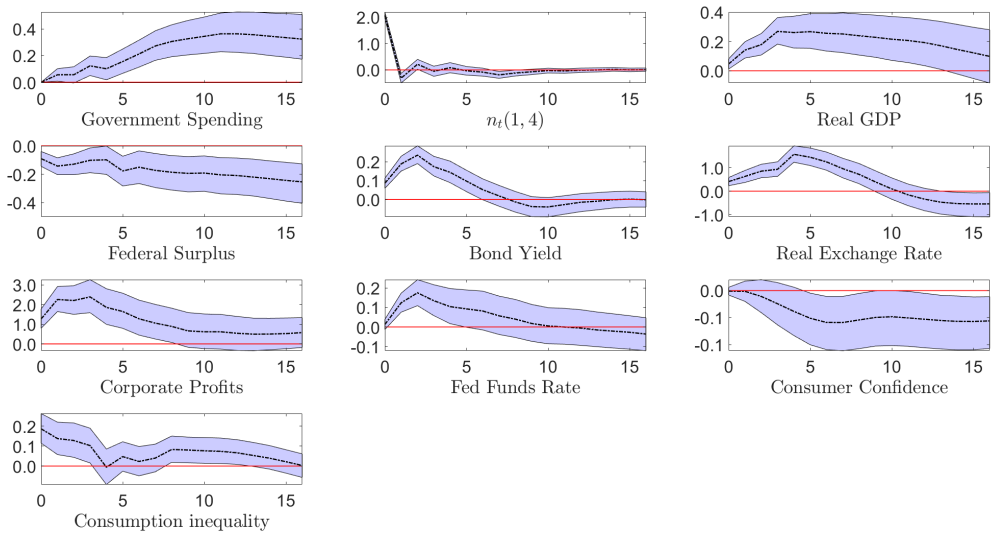

Figure D.9: IRFs of macroeconomic variables to a news shock to government spending.
Notes: solid line is the point-wise posterior median response. Shaded bands are 68% equal-tailed credibility intervals.

D.2 Gini coefficient

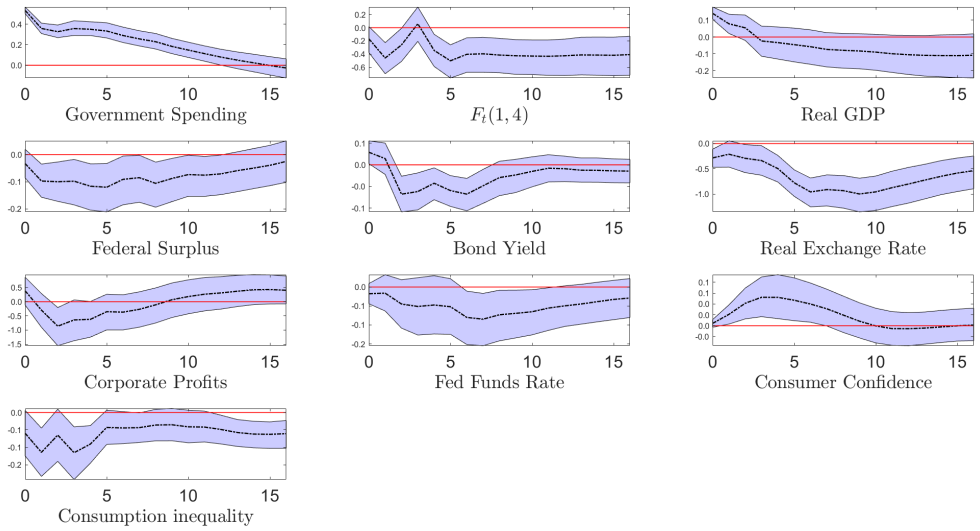

Figure D.10: IRFs of consumption inequality to a surprise shock to government spending.
Notes: solid line is the point-wise posterior median response. Shaded bands are 68% equal-tailed credibility intervals.

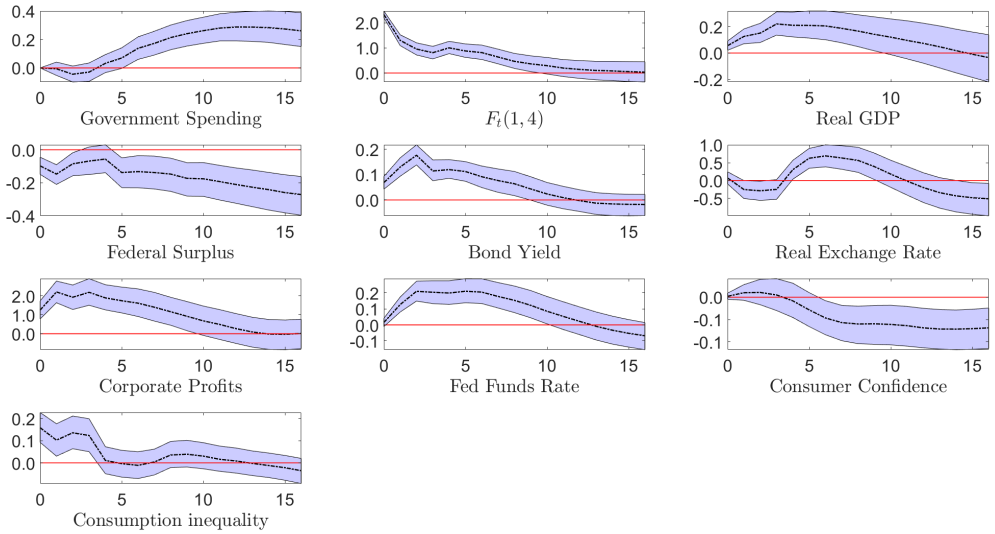

Figure D.11: IRFs of consumption inequality to a news shock to government spending.
Notes: solid line is the point-wise posterior median response. Shaded bands are 68% equal-tailed credibility intervals.

D.3 90*th* - 10*th* percentile

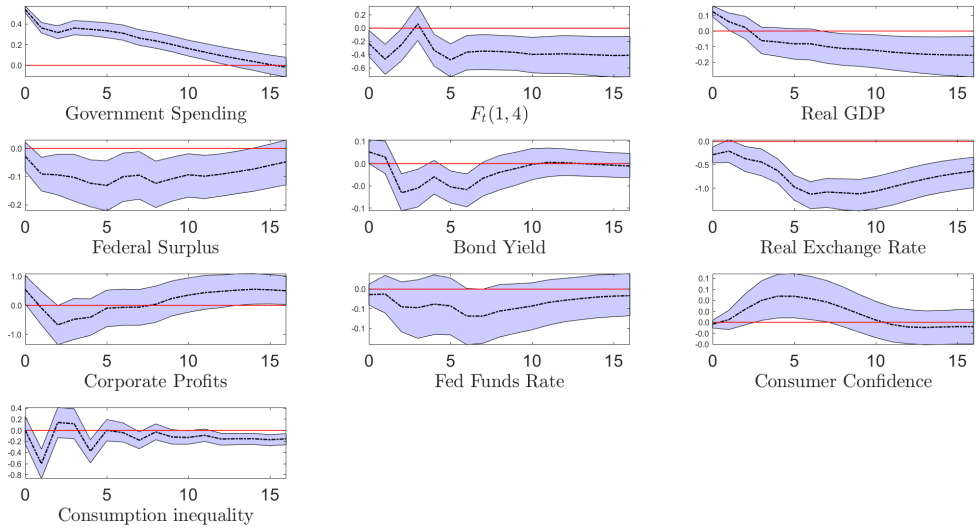

Figure D.12: IRFs of consumption inequality to a surprise shock to government spending.
Notes: solid line is the point-wise posterior median response. Shaded bands are 68% equal-tailed credibility intervals.

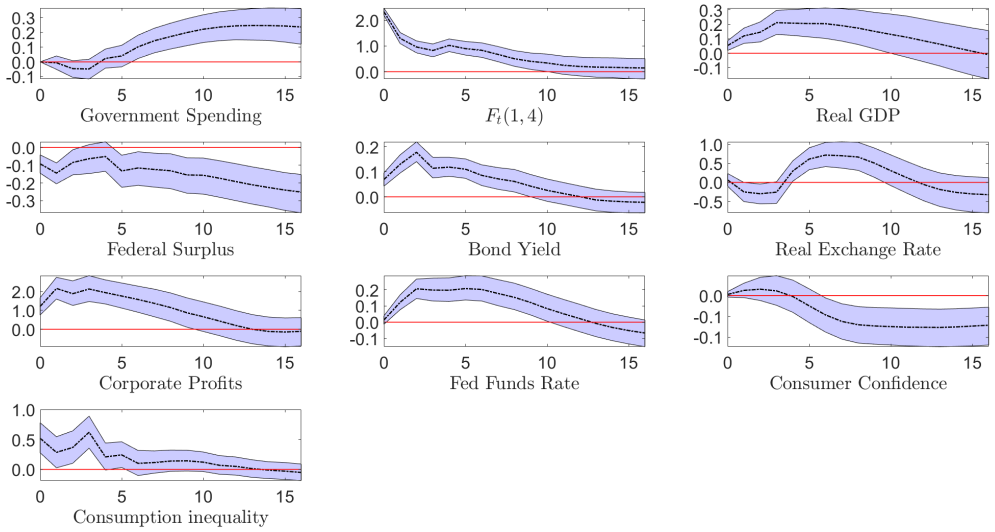

Figure D.13: IRFs of consumption inequality to a news shock to government spending.
Notes: solid line is the point-wise posterior median response. Shaded bands are 68% equal-tailed credibility intervals.

E FAVAR

We show that a FAVAR yields similar results to those obtained from our benchmark VAR. We start from a small-scale VAR containing: (i) real federal government consumption expenditures and gross investment; (ii) real gross domestic product (GDP); (iii) the federal budget surplus as a share of nominal GDP; (iv) the 10-year Treasury constant maturity rate. Table E.4 reveals that the model is not informationally sufficient, as both shocks are predicted by several of the common factors. We thus ensure informational sufficiency by augmenting it with the first five factors from FRED-QD (Table E.5). The IRFs from the FAVAR (Figure E.14 and E.15) largely corroborate the results obtained from the medium-scale model used in Section 2.2.

Surprise shock

PC	Lag 1	Lag 1:2	Lag 1:3	Lag 1:4
1	0.74	0.78	0.92	0.97
2	0.08*	0.13	0.20	0.33
3	0.92	0.70	0.87	0.94
4	0.87	0.14	0.19	0.21
5	0.86	0.97	0.34	0.08*
6	0.34	0.56	0.01**	0.03**
7	0.86	0.82	0.90	0.79

News shock

PC	Lag 1	Lag 1:2	Lag 1:3	Lag 1:4
1	0.98	0.43	0.35	0.44
2	0.10	0.16	0.28	0.12
3	0.19	0.09*	0.15	0.07*
4	0.88	0.91	0.64	0.58
5	0.69	0.53	0.74	0.78
6	0.36	0.28	0.35	0.51
7	0.69	0.92	0.78	0.23

Table E.4: Testing for informational sufficiency (small-scale VAR), * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Surprise shock

PC	Lag 1	Lag 1:2	Lag 1:3	Lag 1:4
1	-	-	-	-
2	-	-	-	-
3	-	-	-	-
4	-	-	-	-
5	-	-	-	-
6	0.71	0.49	0.23	0.33
7	0.99	0.63	0.82	0.74

News shock

PC	Lag 1	Lag 1:2	Lag 1:3	Lag 1:4
1	-	-	-	-
2	-	-	-	-
3	-	-	-	-
4	-	-	-	-
5	-	-	-	-
6	0.44	0.59	0.75	0.88
7	0.56	0.71	0.85	0.78

Table E.5: Testing for informational sufficiency (small-scale VAR + 5 PC), * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

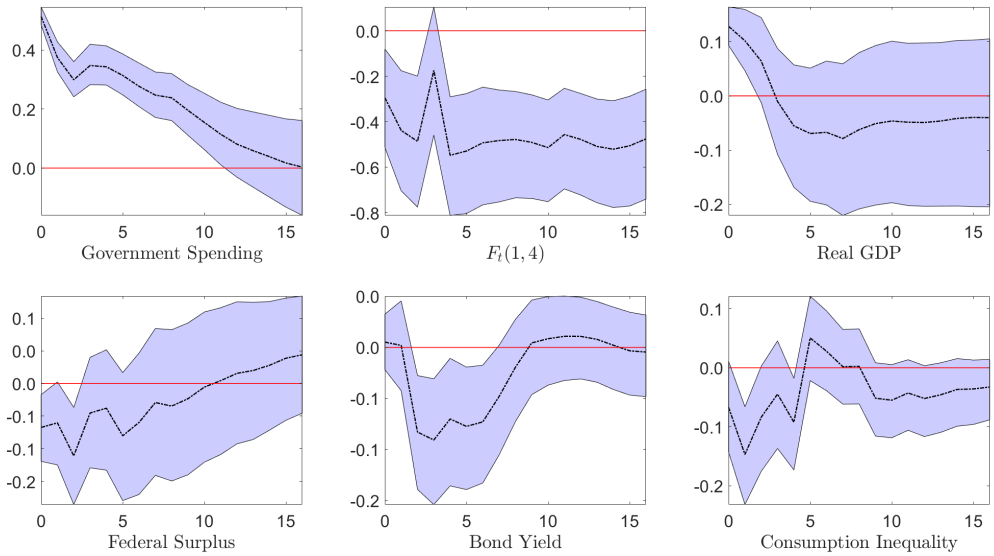

Figure E.14: IRFs to a surprise shock to government spending.
Notes: solid line is the point-wise posterior median response. Shaded bands are 68% equal-tailed credibility intervals.

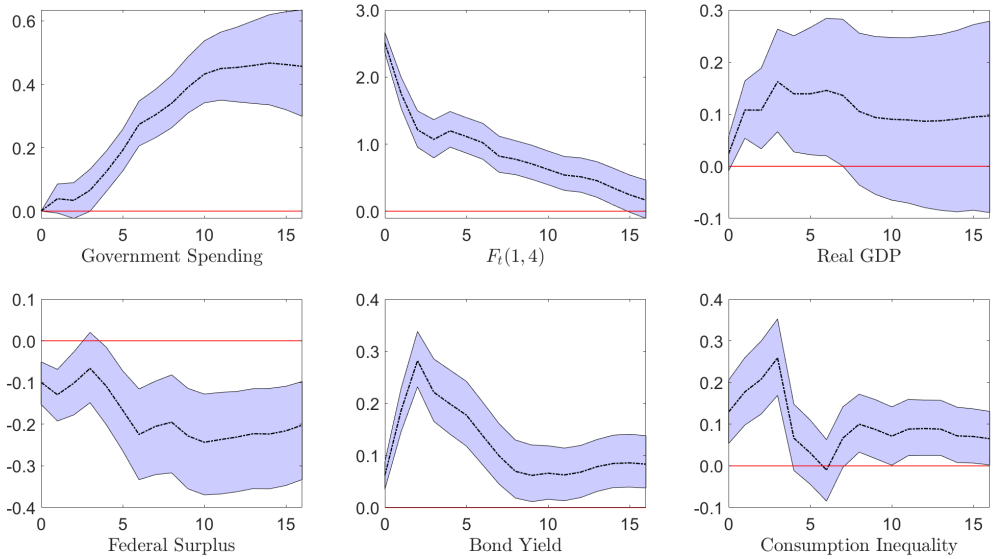

Figure E.15: IRFs to a news shock to government spending.
Notes: solid line is the point-wise posterior median response. Shaded bands are 68% equal-tailed credibility intervals.